Beyond Bundled Aggregators: A Layered Evaluation Framework for Trimming and Client Weighting in Federated IoT Detection

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ABSTRACT Federated learning lets IoT devices train a shared botnet detector without sending raw traffic to a server. Aggregation rules are usually tested as complete bundles. Coordinate-wise trimming removes the lowest and highest client values separately on each parameter coordinate; therefore, the clients removed can differ across coordinates. It is often compared with FedAvg, and FedAvg also weights clients by training-set size, so a difference cannot be traced to either choice. When every client is honest, an extreme coordinate may come from a device whose attack classes differ from the others. This study evaluates trimming and client-size weighting separately. The evaluation has five parts: a two-by-two comparison; a preregistered seven-seed test of Final macro-F1 at round 30 with Holm-corrected paired-t tests; a post-hoc exact sign-flip check; a comparison of the trimmed mean with a coordinate-wise median; and audits of duplicate test rows and client imbalance. A multilayer perceptron and a convolutional network are applied to CIC-IoT2023 at Dirichlet $\alpha = 0.1$ and to a nine-device N-BaIoT partition. Holm correction supports one of 16 contrasts. On the CIC convolutional network, size-weighted FedAvg finishes 10.7252 percentage points below unweighted FedAvg (unadjusted 95% interval $[-15.9527, -5.4977]$; adjusted $p = 0.0384533$). The other 15 contrasts are inconclusive. The median tests and the sign-flip check support no further contrast after Holm correction. Exact training-row copies account for 31.64% of CIC test rows and 59.72% of N-BaIoT test rows. The clients are simulated and honest, and the results depend on the dataset and the client construction.

INDEX TERMS Federated learning, IoT botnet detection, honest heterogeneity, coordinate-wise trimmed mean, client-size weighting, N-BaIoT, CIC-IoT2023

I. INTRODUCTION

Federated learning (FL) can combine client-trained models without storing their raw training records at the server [1]. In IoT intrusion detection, clients can have different traffic and attack-class mixtures. An update at the edge of the client-update distribution does not have to be malicious: it can come from an honest device whose observed classes differ from those of other clients. Whether discarding such coordinates helps the final detector is an empirical question. That question is separate from the Byzantine threat model, in which arbitrary client updates are possible [8].

The original three-arm comparison of FedAvg, FedProx, and an unweighted coordinate-wise trimmed mean could not show that a performance difference was due to trimming alone, because FedAvg also uses client-size weights. Later non-IID experiments added unweighted FedAvg and a size-weighted trimmed mean, forming a 2×2 comparison. The earlier full-scaler runs and the later train-only-scaler experiments are separate lineages. Their scores are not pooled or tested as one family.

This article reports the completed train-only-scaled blocks. CIC-IoT2023 provides a preregistered Dirichlet-label-skew comparison at $\alpha \in \{0.5, 0.1\}$, and N-BaIoT provides a preregistered nine-device partition. A separate seven-seed family then retests only the CIC $\alpha = 0.1$ cells and the N-BaIoT device cells. The comparison asks whether the directions of the trimming and weighting contrasts seen on CIC also appear on the N-BaIoT device clients. It does not say that changing the partition alone caused a difference between datasets. The three reported endpoints are Final accuracy, Final macro-F1, and the mean of 30 round-wise macro-F1 scores. The prespecified directional rules use Final macro-

F1. The round-wise mean is sometimes called "AUC" in the result files. It is not ROC-AUC.

The completed evidence is reported as a five-part evaluation. The parts were not written down as one protocol before every analysis. Parts 1, 2, and 4 were fixed before their own results were used. Parts 3 and 5 were added later from stored results and stored matrices.

1) Part 1 crosses coordinate-wise trimming with client-size weighting in a two-by-two comparison. On the three-seed grids it also asks whether the CIC directional pattern appears on the N-BaIoT device partition. That comparison does not isolate client construction, because the dataset changes as well.
2) Part 2 is the preregistered seven-seed test of Final macro-F1 at round 30, with Holm correction across 16 paired-t tests.
3) Part 3 is a post-hoc exact sign-flip check of those same 16 contrasts. It is not the preregistered decision.
4) Part 4 compares the trimmed mean with a coordinate-wise median on the same four blocks and seven seeds. This comparison was preregistered separately, after the two-by-two study.
5) Part 5 audits duplicate test rows and client imbalance in the stored matrices. No new model was trained for these audits.

The contribution is this separation, reported with those limits. Once trimming and client-size weighting are crossed, apparent benefits depend on the dataset and on how the clients are built. One weighting contrast survives the preregistered seven-seed test. The broader trimming hypotheses and the median comparison remain inconclusive. The clients are simulated partitions of two public corpora. The study includes no privacy mechanism, no rare-class benefit, no adversarial client, and no physical deployment, and it uses no equivalence procedure.

Figure 1 shows the five parts. Parts 1, 2, and 4 were specified before their own results were used. Parts 3 and 5 were added later. The left-to-right order is a reading order, not a causal chain.

II. RELATED WORK AND SCOPE

This article uses a layered evaluation framework for one narrow question: what changes when coordinate-wise trimming and client-size weighting are separated, and how far those comparisons can be trusted. It is not a new detector and not a claim that every robust-aggregation paper should use the same five parts. The related studies below each answer a different question. Table 1 states what Part 1, the two-by-two comparison, adds to each of them.

FedAvg averages locally trained models with client-data-size weights [1]. FedProx adds a proximal term to the local objective [2]. SCAFFOLD uses control variates to correct client drift [3]. FedNova normalizes heterogeneous local progress to address objective inconsistency [4]. These methods act at different points in optimization. SCAFFOLD and FedNova are relevant to heterogeneity, but neither is an arm in the experiments below, and neither replaces the two-factor test of server-side trimming and client-size weighting.

The type and degree of non-IID partitioning can change decentralized-learning accuracy [5], and a broad comparison found no single evaluated FL method that was best in every non-IID case [7]. Label-distribution-skew methods such as FedLC target local minority and missing classes [6]. Those findings motivate class-wise measurement. They do not show that any rare class improved in these runs. Coordinate-wise trimmed means address a different problem: their Byzantine analyses allow arbitrary or adversarial worker behavior [8]. A guarantee under that threat model does not transfer to honest device clients, or to this trimmed-parameter implementation, unless its assumptions are checked.

Recent Byzantine-robust work with label skew has examined the tension between heterogeneous honest gradients and defenses built around a majority of similar updates [9]. The experiments below inject no malicious clients. They ask whether a fixed coordinate-wise trim changes class-balanced detection when all nine simulated clients follow the same training code, and whether client-size weighting changes that answer.

The coordinate-wise median is not the geometric median. Pillutla, Kakade, and Harchaoui aggregate client updates with a geometric median in order to limit corrupted updates [14]. That aggregator is not used here. Cambus, Melnyk, Milentijevic, and Schmid study the coordinate-wise median under Byzantine clients [15]. With nine clients, the median in this article is the fifth ordered value on each coordinate, and clients are not weighted by sample size. Yin et al. give statistical rates for both the coordinate-wise median and the coordinate-wise trimmed mean in a Byzantine model [8]. Those rates are not a result about the honest clients below.

Client weights also come in more than one form. Nguyen, Wu, and Chang reweight individual samples so that a client's distribution moves toward a global distribution [16]. Wilbik, Pękala, Dyczkowski, and Szkoła compare client weights that depend on local model quality [17]. The size weights in this article multiply an entire client parameter vector by that client's training-set size. Sample weights and quality weights are not arms.

Federated IoT anomaly and botnet detection has earlier examples [10], [11]. The N-BaIoT and CIC-IoT2023 source studies establish the two traffic corpora [12], [13]. Albanbay et al. compare model families and local data volume for federated intrusion detection on CICIoT2023, including a convolutional network, under a simulated device population [18]. Their question is model and data scale. Part 1 of this framework instead crosses trimming with client-size weighting on processed versions of both corpora. Parts 2 and 4 add the preregistered multiplicity control and the median comparison. Parts 3 and 5 are later checks on the same stored results. None of those parts is a claim to be the first study of robust aggregation, client weighting, or federated IoT detection.

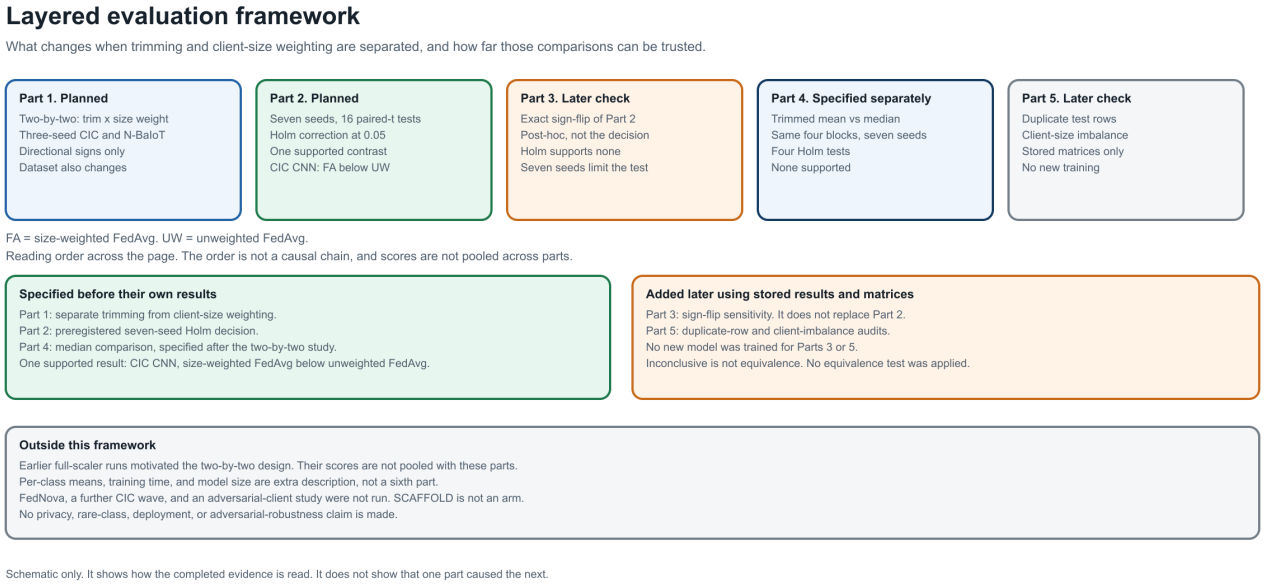


FIGURE 1. The five-part evaluation. Parts 1, 2, and 4 were specified before their own results were used. Parts 3 and 5 were later checks using stored results and stored matrices. Full-scaler runs, per-class means, training time, and model size sit outside the five parts. The left-to-right order is not a causal chain.

TABLE 1. Closest related studies, read against Part 1 of this evaluation. Each row names the question that study asks. The last column is what the two-by-two comparison adds. Parts 2 through 5 are not separate rows in this table.

Study	Question in that study	Increment from this 2×2
Yin *et al.* [8]	Byzantine rates for coordinate-wise median and trimmed mean	Honest clients only. Trimming is crossed with client-size weights.
Pillutla *et al.* [14]	Geometric median under corrupted updates	A different aggregator. The geometric median is not an arm.
Cambus *et al.* [15]	Coordinate-wise median with Byzantine clients	Nine honest clients. The rule is the fifth ordered value. The comparison with the trimmed mean is a separate four-test family.
Nguyen *et al.* [16]	Sample weights for distribution skew	Weights here scale whole client vectors by training-set size.
Wilbik *et al.* [17]	Client weights from local model quality	Size weights are crossed with trimming. Quality weights are not an arm.
Bao *et al.* [9]	Byzantine robustness under label skew	No malicious client is injected.
Albanbay *et al.* [18]	Model and data scale for federated detection on CICIoT2023	Aggregation factorial on processed CIC-IoT2023 and N-BaIoT.
DÏoT and Popoola *et al.* [10], [11]	Federated IoT anomaly and botnet detection	Controlled aggregator comparison on two existing corpora.

III. METHODS

A. HOW THE FIVE PARTS WERE SPECIFIED

Parts 1 and 2 use the same four arms. Part 1 records the three-seed directional comparisons. Part 2 applies the preregistered Holm rule to seven new seeds on four blocks only. Part 4 uses the trimmed-mean cells from Part 2 and adds the coordinate-wise median. Those three parts were specified before their decision results were used. Part 3 applies an exact sign-flip test to the Part 2 differences after those paired-t results existed. Part 5 reads the stored feature matrices and the stored client partitions. It does not add a training run. The sections below give the shared experimental settings, then the rule used by each part.

B. DATA, PARTITIONS, AND PREPROCESSING LINEAGES

N-BaIoT contains traffic from nine commercial IoT device identities [12]. The task here has 11 classes and 115 numeric features. CIC-IoT2023 adds a separate eight-category task with 39 numeric flow features [13]. Table 2 summarizes the processed caches. Each source file is capped at 20,000 rows. A stratified split with seed 42 produces pooled train and test sets. For the two blocks below, a standard scaler is fit on the pooled training matrix only and then applied to train and test. That choice avoids fitting the scaler on test records. It is not fully local or privacy-preserving preprocessing, because the scaler uses pooled training data.

TABLE 2. The two separate train-only-scaled experiment blocks. Counts refer to the processed, capped caches, not the original source datasets. Neither block tests deployment.

Block	Train	Test	Features	Classes	Clients	Client construction
CIC-IoT2023	4,825,692	1,206,424	39	8	9	Dirichlet label allocation
N-BaIoT	1,418,112	354,529	115	11	9	Source device identity

The CIC confirmatory block partitions labels among nine simulated clients at $\alpha = 0.5$ and 0.1. It uses two architectures, four aggregation arms, and seeds $\{42, 123, 7\}$, giving 48 cells. A separate $\alpha = 1.0$ smoke run is not an additional ranking campaign. The N-BaIoT block assigns each training record to its source device, giving nine non-empty, disjoint clients and 24 model-by-arm-by-seed cells. The pooled test split contains records from all nine identities. It measures held-out records from participating devices, not generalization to unseen devices. Seven N-BaIoT training clients contain all 11 classes. The Ennio doorbell and the Samsung webcam each contain six classes and have no Mirai examples in this corpus. That coverage difference is not a measured per-class result.

The earlier full-scaler tables used `scale_on=full`. They remain historical context and are not combined numerically with the `scale_on=train` CIC and N-BaIoT tables. A change between the two tracks cannot be attributed to an aggregator without also accounting for preprocessing and the other design differences.

C. MODELS AND FOUR AGGREGATION ARMS

The MLP has hidden widths 128, 64, 64, and 32, with ReLU activations and layer normalization. Its first three hidden blocks use dropout rates 0.2, 0.2, and 0.1. The one-dimensional CNN treats each feature vector as a sequence. It uses two convolutional layers, first 1 to 32 channels with kernel 5 and then 32 to 64 channels with kernel 3, followed by adaptive average pooling and a 64-to-32 classifier head. The same implementations are used for all four arms in a block. They are not architectures tuned separately for each aggregator. All clients take part in each of 30 communication rounds. Each client trains for three local epochs with Adam at learning rate 10^{-3} and batch size 512, then returns its parameters. The server scores the aggregated model on the pooled holdout after each round. Each four-arm comparison uses the same architecture and seed.

Table 3 displays the factorial interpretation. Let w_k denote a returned client parameter vector, n_k its training-set size, and $n = \sum_{k=1}^{K} n_k$. The untrimmed arms are size-weighted FedAvg (FA) and unweighted FedAvg (UW):

$$w_{\mathrm{FA}} = \sum_{k=1}^{K} \frac{n_k}{n} w_k, \qquad w_{\mathrm{UW}} = \frac{1}{K} \sum_{k=1}^{K} w_k. \tag{1}$$

For each parameter coordinate, the trimmed arms sort the nine returned values and discard one value at each tail. FedTrimmedAvg (FT) averages the seven retained values equally; WeightedFedTrimmedAvg (WT) uses the retained clients' sample-size weights, normalized over those seven values. Thus the configured trimming fraction 0.1 removes $1/9$ of the clients at each tail for this $K = 9$ implementation. FedProx belongs to a separate, scoped optimizer/regularization investigation and is not a fifth arm of this factorial comparison.

TABLE 3. The four aggregation arms as a two-factor design.

Arm	Coordinate trim	Size weights
UW	No	No
FA	No	Yes
FT	Yes	No
WT	Yes	Yes

More formally, let S_j be the client indices retained for coordinate j after removing the lowest and highest returned values. The two trimmed arms are

$$w_{\mathrm{FT},j} = \frac{1}{|S_j|} \sum_{k \in S_j} w_{k,j}, \qquad w_{\mathrm{WT},j} = \frac{\sum_{k \in S_j} n_k w_{k,j}}{\sum_{k \in S_j} n_k}. \tag{2}$$

Because S_j is coordinate-specific, WT is not obtained by selecting seven whole client models once per round; the survivor set can differ across parameters.

D. ENDPOINTS AND PRESPECIFIED DIRECTIONAL RULES

The reported endpoints are Final accuracy, Final macro-F1, and the mean of round-wise macro-F1 over rounds 1–30. Best-round scores are secondary diagnostics. They do not replace Final. Macro-F1 is the average of the per-class F1 values, each formed from that class's precision and recall. For each model and each CIC α cell, the preregistered contrasts are FT $-$ UW and WT $-$ FA for adding trimming, and FA $-$ UW and WT $-$ FT for adding size weighting. Trim-help requires both mean trim contrasts to be positive. Weight-hurt requires both mean weight contrasts to be negative. The N-BaIoT device block uses the same pair definitions by model. It has no preregistered FT-first ranking hypothesis. The three-seed directional results use unrounded stored Final macro-F1 means over three matched seeds. An exact tie does not meet the descriptive rule. The word "supported" is reserved for the seven-seed Holm decision. Paired seed signs are reported with the means. There is no $n = 3$ significance or equivalence claim. The seven-seed family in Section IV-E uses a different preregistered rule: 16 two-sided paired t tests on seeds $\{127, 131, 137, 139, 149, 151, 157\}$, with Holm correction across all 16 tests at 0.05. That rule does not reclassify the three-seed tables.

E. INTEGRITY AND REPORTING CONTROLS

Each completed row is bound to the processed train-scaler cache and the metadata hashes recorded for its block. Histories contain exactly 30 rounds, and the stored Final macro-F1 is checked against round 30 before a cell counts as complete. The CIC and N-BaIoT manifests cover the processed caches and the written histories. They do not independently hash every original source CSV. Final accuracy and Final macro-F1 use round 30. They are not replaced by the maximum score over the 30 pooled-test evaluations. That rule keeps a test-selected best round from replacing the prespecified final model.

IV. RESULTS

The results follow the five parts named in the Introduction. Parts 1, 2, and 4 are the planned comparisons. Parts 3 and 5 are later checks.

A. PART 1: TWO-BY-TWO COMPARISON ON CIC-IOT2023

Table 4 gives three-seed means of the three endpoints. Under the prespecified Final macro-F1 rule, FT has the highest arm mean, both trim contrasts have positive means, and both weight contrasts have negative means in each of the four model-by-α cells. The descriptive directional rule is met in all four cells. That does not mean every seed has the same sign, and it does not mean a significance test was passed.

B. PART 1 CONTINUED: TWO-BY-TWO COMPARISON ON THE N-BAIOT DEVICE PARTITION

Table 5 reports the corresponding device-partition means for all three primary endpoints. Neither descriptive directional rule was met for either model: both N-BaIoT scores are $0/2$. FT minus UW is negative in all three seeds in both models, whereas WT minus FT is positive in all three seeds in both models. The remaining contrast signs vary by seed. WT has the highest descriptive MLP Final macro-F1 mean; UW exceeds WT by only about 0.22 percentage points in the CNN Final macro-F1 means. There is no preregistered FT-first decision for this block, and these rankings should not be treated as superiority or equivalence findings.

C. PART 1 CONTINUED: PAIRED CONTRASTS

Table 6 exposes the paired Final macro-F1 contrasts used for the directional decisions. Values are mean percentage-point differences over the three matched seeds. Parentheses give the number of seeds with the prespecified direction: positive for the two trim contrasts and negative for the two weight contrasts. The CIC descriptive rule uses the signs of the unrounded three-seed means; the table makes clear where an individual seed disagrees. On N-BaIoT, neither descriptive directional rule was met.

Figure 2 complements Table 6 by displaying every seed-level paired difference as well as its three-seed mean. It prevents a positive mean from being mistaken for unanimous agreement across the three seeds and makes the cross-setting sign reversal visible without combining raw performance levels.

The CIC and N-BaIoT directional outcomes differ, but the datasets and client partitions also differ. Their contrast signs can be discussed side by side; their raw scores are not pooled, and this study cannot isolate device partitioning as the cause of the difference. The 30-round histories also show that some CNN cells end substantially below their own best round. We therefore preserve the prespecified Final endpoint and report Best only as a secondary trajectory diagnostic.

D. PART 1 DESCRIPTIVE INTERACTION

The interaction $(\mathrm{WT} - \mathrm{FA}) - (\mathrm{FT} - \mathrm{UW})$ stays inside Part 1. It describes the crossing of trimming and client-size weighting. It was prespecified as a descriptive summary, it is outside the 16-test Holm family, and no interaction p-value is used as a decision. It is not a sixth part of the framework. Table 7 and Fig. 3 report the seven-seed mean, the unadjusted 95% paired-t interval, and the positive, tie, and negative seed counts. On T1 and T2 the interval excludes zero. On T5 and T6 it includes zero. These intervals describe the estimates. They do not add a supported comparison, and they do not change Table 8.

E. PART 2: PREREGISTERED SEVEN-SEED TEST

The three-seed boards above do not include a multiplicity-controlled test. That test was preregistered separately and is now closed. It uses the same train-only scaler, nine simulated clients, 30 rounds, and four arms, but only four blocks: CIC-IoT2023 MLP and CNN at Dirichlet $\alpha = 0.1$ (T1 and T2), and the fixed N-BaIoT nine-device partition for MLP and CNN (T5 and T6). The CIC $\alpha = 0.5$ cells are not in this family. On CIC, each new seed sets matched random conditions for all four arms. On N-BaIoT, the device partition stays fixed across seeds. The seven prospective seeds are $\{127, 131, 137, 139, 149, 151, 157\}$. The three earlier seeds are excluded from the tests. One hundred twelve new cells were completed. The endpoint is unrounded Final macro-F1 at round 30.

The family contains 16 two-sided paired comparisons: $\mathrm{FT} - \mathrm{UW}$ and $\mathrm{WT} - \mathrm{FA}$ for trimming, and $\mathrm{FA} - \mathrm{UW}$ and $\mathrm{WT} - \mathrm{FT}$ for size weighting, in each of the four blocks. Holm adjustment is across all 16 comparisons. A trim-help block is supported only when both trim comparisons are significant in the positive direction. A weight-hurt block is supported only when both weight comparisons are significant in the negative direction. Any other significant direction is a reversal for that comparison. Otherwise the comparison is inconclusive. Inconclusive does not mean the arms are equivalent, and no equivalence test was run.

Table 8 reports every comparison. Means and intervals are shown to four decimal places, and p-values use up to six significant digits. Only one comparison meets the decision rule. On T2, the CIC CNN, $\mathrm{FA} - \mathrm{UW}$ has mean -10.7252 percentage points, unadjusted 95% paired-t interval $[-15.9527, -5.4977]$, signs $0/0/7$, raw $p = 0.00240333$,

TABLE 4. CIC-IoT2023 train-only-scaled confirmatory grid: three-seed means of the three primary endpoints. "Round mean F1" is the mean of 30 round-wise macro-F1 values, not ROC-AUC. FA, UW, FT, and WT are defined in Methods. Directional decisions use unrounded Final macro-F1, not the displayed five-decimal means.

Model	α	Arm	Final Acc	Final F1	Round mean F1
MLP	0.5	FA	0.88929	0.61594	0.59854
		UW	0.89461	0.66156	0.64397
		FT	0.89369	0.67497	0.65881
		WT	0.89484	0.66704	0.65075
MLP	0.1	FA	0.85784	0.42598	0.42049
		UW	0.83488	0.55188	0.53443
		FT	0.83486	0.59686	0.58791
		WT	0.82842	0.50295	0.48721
CNN	0.5	FA	0.88626	0.57053	0.53056
		UW	0.89250	0.63050	0.58582
		FT	0.89232	0.63993	0.59024
		WT	0.89362	0.63511	0.58531
CNN	0.1	FA	0.85276	0.40552	0.38295
		UW	0.86244	0.46235	0.43128
		FT	0.83198	0.52651	0.46655
		WT	0.78348	0.43521	0.37436

TABLE 5. N-BaIoT nine-device partition: three-seed means of the three primary endpoints on a pooled holdout containing all nine device identities. "Round mean F1" is not ROC-AUC. This is not an unseen-device test.

Model	Arm	Final Acc	Final F1	Round mean F1
MLP	FA	0.82616	0.79348	0.76527
	UW	0.81682	0.78192	0.75949
	FT	0.81229	0.77297	0.76033
	WT	0.83085	0.79982	0.76512
CNN	FA	0.80437	0.75374	0.73127
	UW	0.81845	0.77796	0.72380
	FT	0.79545	0.73851	0.73331
	WT	0.81722	0.77580	0.72995

TABLE 6. Paired Final macro-F1 contrasts in percentage points, with desired-direction seed counts in parentheses. Required directions are $+$ for FT$-$UW and WT$-$FA, and $-$ for FA$-$UW and WT$-$FT.

Block	Cell	FT$-$UW $(+)$	WT$-$FA $(+)$	FA$-$UW $(-)$	WT$-$FT $(-)$
CIC	MLP, $\alpha = 0.5$	+1.3405 (3/3)	+5.1101 (3/3)	$-$4.5623 (3/3)	$-$0.7927 (3/3)
CIC	MLP, $\alpha = 0.1$	+4.4978 (3/3)	+7.6971 (2/3)	$-$12.5899 (3/3)	$-$9.3906 (3/3)
CIC	CNN, $\alpha = 0.5$	+0.9438 (2/3)	+6.4580 (3/3)	$-$5.9970 (3/3)	$-$0.4828 (2/3)
CIC	CNN, $\alpha = 0.1$	+6.4167 (3/3)	+2.9693 (2/3)	$-$5.6830 (3/3)	$-$9.1304 (3/3)
N-BaIoT	MLP, device	$-$0.8952 (0/3)	+0.6333 (2/3)	+1.1559 (1/3)	+2.6844 (0/3)
N-BaIoT	CNN, device	$-$3.9444 (0/3)	+2.2054 (2/3)	$-$2.4216 (2/3)	+3.7283 (0/3)

TABLE 7. Part 1. Descriptive trimming-by-weighting interaction on the seven prospective seeds, in Final macro-F1 percentage points. Signs are positive, tie, and negative counts. No decision p-value is attached.

Block	Mean	95% interval	Signs
T1	7.0612	$[0.3531,\ 13.7692]$	7/0/0
T2	7.8498	$[1.8981,\ 13.8015]$	6/0/1
T5	-1.6837	$[-4.7734,\ 1.4060]$	3/0/4
T6	0.5505	$[-8.9670,\ 10.0680]$	2/0/5

and Holm-adjusted $p = 0.0384533$. Size weighting is associated with a lower Final macro-F1 than equal client weights in that one block. The other 15 comparisons are inconclusive. There is no reversal.

Neither complete hypothesis is supported. Trim-help requires both trim comparisons in a block to be supported in the positive direction, and no block meets that requirement. Weight-hurt requires both weight comparisons in a block to be supported in the negative direction, and no block meets that requirement. The supported T2 FA $-$ UW result is one of the two weight components in that block. T2 WT $-$ FT is inconclusive, with Holm-adjusted $p = 1$. Ten-seed summaries were prespecified as descriptive and are not used to override Table 8.

F. PART 3: POST-HOC EXACT SIGN-FLIP CHECK

The paired-t test at $n = 7$ leans on a normal approximation for the mean of seven differences. A post-hoc exact sign-flip test uses the same seven differences and makes a weaker assumption: under exchangeability of the sign of each paired difference, all $2^7 = 128$ sign assignments are equally likely. The statistic is the absolute sum of the signed differences. The two-sided p-value is the fraction of the 128 assignments

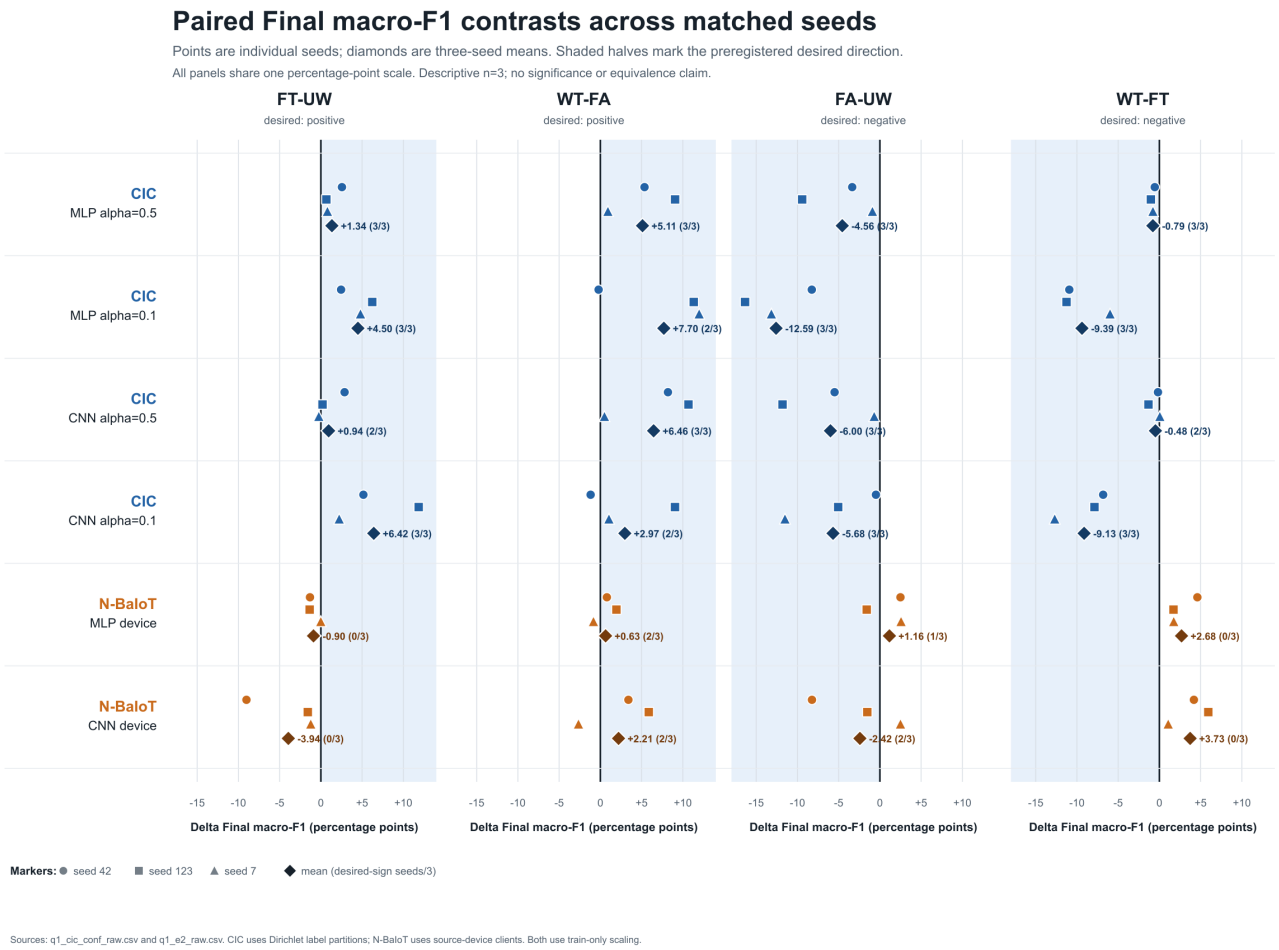


FIGURE 2. Seed-level paired Final macro-F1 contrasts for the two trim and two weight comparisons. Small markers are the three matched seeds and diamonds are their means; the lightly shaded half of each panel is the prespecified desired direction. Counts report seeds in that direction. CIC and N-BaIoT are displayed together for sign-pattern comparison only; their scores are not pooled. All summaries are descriptive at $n = 3$.

TABLE 8. Part 2, the seven-seed family. Paired Final macro-F1 differences, in percentage points. The interval is the unadjusted 95% paired-t confidence interval. Signs are positive, tie, and negative counts. Holm adjustment uses all 16 rows. Only T2 $\mathrm{FA} - \mathrm{UW}$ is supported. This paired-t Holm family is the preregistered decision.

Block	Contrast	Mean	95% CI	Signs	Raw p	Holm p	Decision
T1	FT−UW	0.3714	[−4.2080, 4.9507]	5/0/2	0.849257	1	inconclusive
T1	WT−FA	7.4325	[2.6661, 12.1990]	7/0/0	0.00880574	0.12328	inconclusive
T1	FA−UW	−12.2101	[−19.7262, −4.6941]	1/0/6	0.00732394	0.109859	inconclusive
T1	WT−FT	−5.1490	[−13.1925, 2.8946]	2/0/5	0.168306	1	inconclusive
T2	FT−UW	−1.5393	[−4.7906, 1.7121]	3/0/4	0.290708	1	inconclusive
T2	WT−FA	6.3105	[1.1477, 11.4734]	6/0/1	0.024293	0.315809	inconclusive
T2	FA−UW	−10.7252	[−15.9527, −5.4977]	0/0/7	0.00240333	0.0384533	supported
T2	WT−FT	−2.8754	[−7.8214, 2.0706]	2/0/5	0.20472	1	inconclusive
T5	FT−UW	−0.0201	[−2.4796, 2.4394]	4/0/3	0.984693	1	inconclusive
T5	WT−FA	−1.7038	[−3.2609, −0.1466]	0/0/7	0.0366684	0.440021	inconclusive
T5	FA−UW	1.6414	[−0.7365, 4.0193]	5/0/2	0.142178	1	inconclusive
T5	WT−FT	−0.0423	[−1.3127, 1.2281]	4/0/3	0.937724	1	inconclusive
T6	FT−UW	0.7091	[−3.2827, 4.7008]	6/0/1	0.679	1	inconclusive
T6	WT−FA	1.2596	[−4.7011, 7.2203]	2/0/5	0.623612	1	inconclusive
T6	FA−UW	−0.4904	[−5.7321, 4.7514]	3/0/4	0.826545	1	inconclusive
T6	WT−FT	0.0601	[−4.2701, 4.3904]	2/0/5	0.974004	1	inconclusive

whose absolute sum is at least the observed absolute sum. Holm adjustment is then applied to these 16 exact p-values.

Table 9 reports that sensitivity. No comparison has a sign-flip Holm-adjusted p-value below 0.25. The smallest raw values are $2/128 = 0.015625$, for T1 $\mathrm{WT} - \mathrm{FA}$, T2 $\mathrm{FA} - \mathrm{UW}$, and T5 $\mathrm{WT} - \mathrm{FA}$. With seven nonzero differences, $2/128$ is the smallest two-sided p-value this enumeration can return. Holm adjustment across 16 tests therefore cannot fall below $16 \times 2/128 = 0.25$. The exact family-wise test cannot resolve this 16-comparison family at level 0.05 when

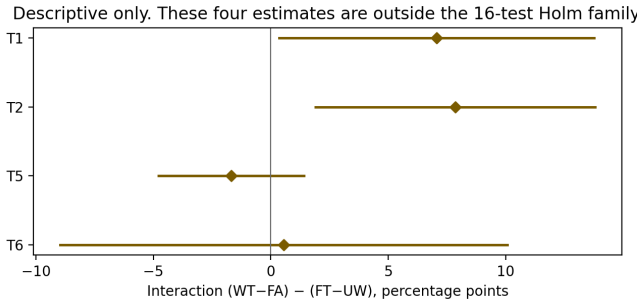


FIGURE 3. Part 1. Descriptive interaction estimates. The bar is the unadjusted 95% paired-t interval. These four estimates are outside the Holm family and are not a sixth part of the evaluation.

TABLE 9. Post-hoc exact sign-flip p-values for the 16 contrasts in Table 8. Holm adjustment uses these 16 exact p-values only. No row is supported at 0.05. This table is a sensitivity analysis, not a replacement for Table 8.

Block	Contrast	Sign-flip p	Holm p
T1	FT−UW	0.875000	1
T1	WT−FA	0.015625	0.250000
T1	FA−UW	0.031250	0.406250
T1	WT−FT	0.171875	1
T2	FT−UW	0.281250	1
T2	WT−FA	0.031250	0.406250
T2	FA−UW	0.015625	0.250000
T2	WT−FT	0.187500	1
T5	FT−UW	0.968750	1
T5	WT−FA	0.015625	0.250000
T5	FA−UW	0.156250	1
T5	WT−FT	0.937500	1
T6	FT−UW	0.515625	1
T6	WT−FA	0.640625	1
T6	FA−UW	0.828125	1
T6	WT−FT	0.968750	1

each contrast has only seven seeds. The preregistered paired-t Holm decision remains the official result, and under that rule T2 FA − UW stays supported. Reviewers should see both statements together: the official test supports one weighting contrast, and the small-sample exact test supports none.

G. PART 4: TRIMMED MEAN AGAINST THE COORDINATE-WISE MEDIAN

Part 4 was specified separately from Part 2. It asks whether the coordinate-wise trimmed mean differs from the coordinate-wise median. The comparison uses the same four blocks and the same seven seeds, $\{127, 131, 137, 139, 149, 151, 157\}$. Each median cell uses Adam at learning rate 10^{-3}, three local epochs, batch size 512, and 30 rounds. On each coordinate the server keeps the fifth ordered value across the nine client updates and does not weight clients by sample size. The comparator is the trimmed-mean cell from the seven-seed family for the same block and seed. Those trimmed-mean cells were not retrained. FedNova is not an arm in this comparison. The supplement records why an earlier plan that included FedNova ended before any decision cell.

The endpoint is unrounded Final macro-F1 at round 30. Each test is a two-sided paired t test on the seven seed-wise differences, trimmed mean minus coordinate-wise median, with $n = 7$ and $\mathrm{df} = 6$. Holm adjustment covers these four tests only, in the order T1, T2, T5, T6, at family-wise level 0.05. A positive supported mean would say the trimmed mean is higher than the median in that block only. A negative supported mean would say the median is higher in that block only. This adjustment is not combined with Table 8.

Table 10 reports every comparison. Means and intervals are in percentage points to four decimal places, and p-values use up to six significant digits. None of the four tests is supported. No equivalence procedure was applied, so the table does not establish that the trimmed mean and the median are interchangeable. The supported T2 FA − UW result in Table 8 is unchanged.

H. PART 5: CLIENT IMBALANCE

Part 5 was computed after the training runs, from the stored partitions. It compares client size and label mix. It does not add a new model. Table 11 uses the same Dirichlet allocations and the same device partition as the seven-seed family. A partition is included only when its hash matches the recorded configuration. Jensen–Shannon divergence is computed in bits between each client's training-label distribution and the pooled training-label distribution. A missing class is a class with zero training rows on that client.

On CIC-IoT2023 at $\alpha = 0.1$, client size is highly unbalanced. The largest-to-smallest client ratio runs from 10.8 to $1{,}555$, the coefficient of variation from 0.73 to 1.63, and the largest client's share of training rows from 0.270 to 0.584. Mean Jensen–Shannon divergence is between 0.357 and 0.498 bits. Every seed leaves at least one class missing from at least one client; the mean number of missing classes per client is about 1.22 to 1.33, and the worst client in a seed misses 2 to 4 classes.

The N-BaIoT device partition is much closer to balance. Its nine sizes run from $95{,}875$ to $176{,}174$ rows, so the largest-to-smallest ratio is 1.84, the coefficient of variation is 0.209, and the largest share is 0.124. Mean Jensen–Shannon divergence is 0.054 bits and the maximum is 0.233 bits. The mean number of missing classes is 1.11, while the worst client misses 5 classes. That worst-client count is consistent with the two devices, the Ennio doorbell and the Samsung webcam, that contain six of the eleven classes and have no Mirai examples in this corpus.

These rows describe the partitions. They do not identify which change, dataset or client construction, produced the difference in directional pattern between CIC-IoT2023 and N-BaIoT.

I. PART 5 CONTINUED: DUPLICATE ROWS AND SOURCE OVERLAP

Both datasets reuse a pooled train/test structure. The leakage check compares stored float32 feature rows by exact equality, which is the matrix the models saw. It does not rebuild the caches. A feature with zero variance can make distinct raw rows identical after scaling, so an exact match is evidence about the stored matrix, not a reconstructed raw-file identity.

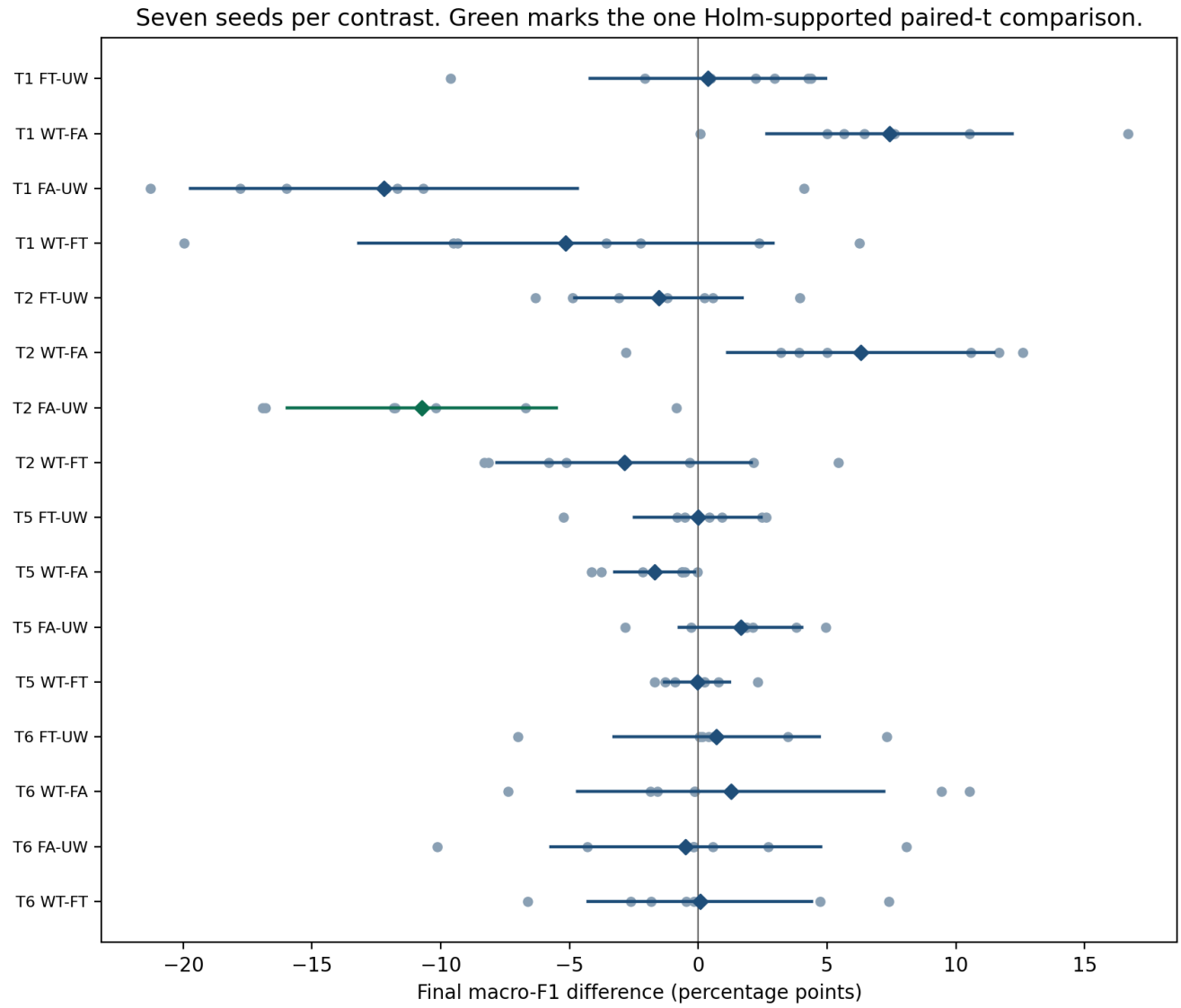


FIGURE 4. All 16 seven-seed contrasts. Dots are the seven paired Final macro-F1 differences, in percentage points. The bar is the unadjusted 95% paired-t interval and the diamond is the mean. The green row is the one comparison with paired-t Holm-adjusted $p < 0.05$.

TABLE 10. Part 4. Paired Final macro-F1 differences, in percentage points, over seven seeds. The difference is the trimmed mean minus the coordinate-wise median. The interval is the unadjusted 95% paired-t confidence interval. Signs are negative, tie, and positive counts, in that order. Holm adjustment uses these four rows only. No row is supported.

Block	Contrast	Mean	95% CI	−/0/+	Raw p	Holm p	Decision
T1	FT−median	0.3471	[−1.6900, 2.3842]	3/0/4	0.691258	1	inconclusive
T2	FT−median	0.6388	[−1.7203, 2.9980]	3/0/4	0.532201	1	inconclusive
T5	FT−median	−1.7913	[−4.5363, 0.9537]	4/0/3	0.161432	0.645727	inconclusive
T6	FT−median	−1.6825	[−4.8882, 1.5233]	3/0/4	0.246421	0.739263	inconclusive

The CIC training matrix has 4,825,692 rows and 3,565,650 unique rows. Of 1,206,424 test rows, 381,765 (31.64%) are exact copies of some training row. All eight classes appear in both splits. The stored cache has no test source identifier, and the stored training source identifier takes only one value, so overlap of CIC source files between train and test cannot be counted from this cache.

The N-BaIoT training matrix has 1,418,112 rows and 742,236 unique rows. Of 354,529 test rows, 211,718 (59.72%) are exact copies of some training row. All eleven classes appear in both splits. Source identifiers 0 through 8 appear in both splits, and all 354,529 test rows come from a source that also appears in training. The N-BaIoT evaluation is a pooled within-device holdout. It is not a test on unseen devices.

TABLE 11. Client-size imbalance, missing-class coverage, and label divergence for the seven-seed partitions. CIC rows are Dirichlet $\alpha = 0.1$. The N-BaIoT row is the fixed device partition. JS is Jensen–Shannon divergence in bits against the pooled training-label distribution. Counts are rounded from the recomputed partition tables.

Partition	Min	Max	Max/min	CV	Largest share	Mean missing	Max missing	Mean JS	Max JS
CIC, seed 127	7,652	2,395,227	313.0	1.39	0.496	1.22	2	0.495	0.804
CIC, seed 131	120,178	1,301,733	10.8	0.73	0.270	1.22	3	0.357	0.685
CIC, seed 137	1,219	1,895,895	1,555.3	1.09	0.393	1.33	3	0.401	0.906
CIC, seed 139	27,236	2,568,302	94.3	1.41	0.532	1.22	3	0.412	0.806
CIC, seed 149	34,483	1,406,112	40.8	1.10	0.291	1.22	3	0.498	0.746
CIC, seed 151	5,335	2,008,503	376.5	1.20	0.416	1.22	4	0.433	0.767
CIC, seed 157	3,147	2,819,296	895.9	1.63	0.584	1.22	3	0.481	0.913
N-BaIoT, device	95,875	176,174	1.84	0.21	0.124	1.11	5	0.054	0.233

J. ADDITIONAL DESCRIPTION, NOT A SIXTH PART: PER-CLASS MEANS

The five parts above are the evaluation named in the abstract. The displays in this subsection and the next two are extra description. They do not add a part and they do not change a decision. Fig. 5 shows mean per-class F1 for the CIC-IoT2023 MLP and CNN at $\alpha = 0.1$, using the three seeds $\{42, 123, 7\}$ and all four arms. This display is descriptive. It is not part of the seven-seed Holm family, and it is not a rare-class finding. Mirai is near the top of the color scale for every arm on both models. Web remains low for every arm on both models. The figure shows that macro-F1 averages classes with very different detection levels. It does not identify which coordinates were trimmed.

K. ADDITIONAL DESCRIPTION: FINAL SCORE, TRAINING TIME, AND MODEL SIZE

These quantities explain the Part 2 endpoint and the cost of the cells. They are not a separate evaluation part. The prespecified endpoint is Final macro-F1 at round 30. Table 12 compares that score with the best round-wise macro-F1 inside the same 30 rounds, for the 112 seven-seed cells. Histories were checked against the seed table. Mean gaps are 1.204, 1.456, 1.697, and 3.242 percentage points on T1, T2, T5, and T6. The largest single-cell gap is 9.019 percentage points, on T6. Only 3, 5, 4, and 6 cells in those blocks have their best score at round 30. Late-round decline is present. The average gap is a few points, and the reported comparison continues to use round 30.

Recorded training time for those 112 cells is the stored `train_s` field, summed within each block and converted to hours. It is not calendar span and it is not a campaign-wide budget. Table 13 gives the mean, maximum, and sum. The four sums add to 171.954 hours. The 28 median cells have a separate elapsed-time record, from each cell's stored start to its stored finish: mean 1.502 hours, maximum 2.323 hours, and sum 42.066 hours. That sum is elapsed wall-clock time for those 28 cells. A different planning quantity is stated in the supplement and is not this sum.

L. OUTSIDE THE FIVE PARTS: DEVELOPMENTAL LINEAGE

The five-part evaluation uses the train-only-scaled blocks. The runs below came earlier, use a different scaler, and are not one of the five parts. Before the train-scaler blocks, reporting was corrected so that round 30, rather than the maximum test-set F1 across 30 rounds, was the primary result. Adding unweighted FedAvg and size-weighted trimmed mean to an older three-arm, full-scaler grid exposed the trim-versus-weight confound and motivated the four-arm design. Those runs remain a separate `scale_on=full` lineage. No unweighted-FedAvg or size-weighted-trimmed-mean IID anchors were run, and their scores are not combined with Tables 4–6.

A scoped FedProx diagnostic belongs to that full-scaler lineage and is not a fifth arm. On its CIC CNN $\alpha = 1.0$, seed-42 cell, $\mu = 0.01$ reduced Final macro-F1 by 6.94 percentage points relative to the matched $\mu = 0$ run under Adam and by 0.72 points under SGD. That cell does not support a general claim about FedProx. Its scores are not combined with the train-scaler tables.

V. CONCEPTUAL BRIDGE FOR PART 1

This section is a mechanism reading of Part 1. It is not a sixth part of the evaluation, and it is not a result from Parts 2 through 5. Coordinate-wise trimmed-mean methods have been studied under Byzantine models in which some workers may send arbitrary or adversarial messages [8]. These experiments have no adversarial clients. Every client follows the same training procedure. A trimmed-mean result under its own threat model does not establish Byzantine robustness for this detector, and it does not by itself predict an advantage under honest label skew.

Two honest mechanisms can move the aggregate. A client with more training rows has more weight in FA than in UW. Clients with different class mixtures can return different parameter vectors after only three local epochs, and those vectors are not solved local risk minimizers. In the device block, two clients have no Mirai training examples. Those differences could place some honest coordinates in the tails that FT or WT exclude. The study has not established which clients are trimmed on which coordinates, or whether that explains the measured contrasts.

Fig. 5 shows that per-class F1 already differs across attack families on the three-seed CIC grid. Macro-F1 averages those levels. The figure is descriptive. It does not show that trimming protects or harms a rare class, and WT − UW changes both factors at once, so it is not read as a one-factor effect.

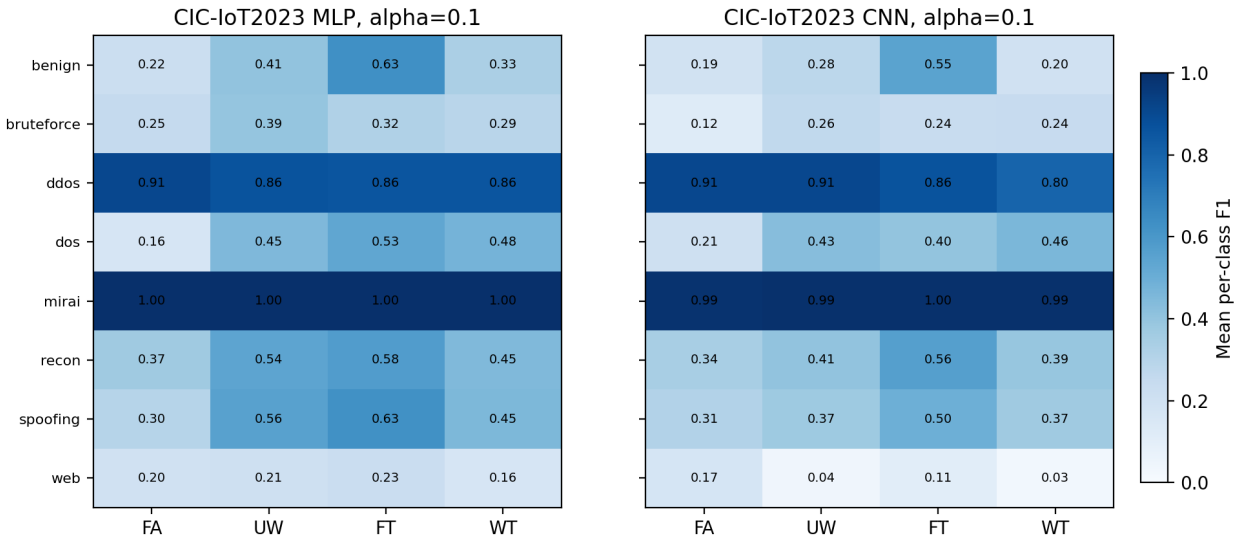


FIGURE 5. Three-seed mean per-class F1 on CIC-IoT2023 at $\alpha = 0.1$. Columns are the four arms. The display is descriptive and is not a Holm test.

TABLE 12. Final versus best round-wise macro-F1 on the 112 seven-seed cells. Gaps are Best minus Final, in percentage points. "Final is best" counts cells whose best round is round 30.

Block	Mean gap	Max gap	Final is best	Median best round	Mean best round
T1	1.204	6.165	3/28	27	24.79
T2	1.456	5.614	5/28	25	24.86
T5	1.697	4.677	4/28	26	23.86
T6	3.242	9.019	6/28	26	22.39

TABLE 13. Recorded training time, in hours, for the 112 seven-seed cells. Parameter counts, including LayerNorm scale and shift, are 20,456 (CIC MLP), 8,808 (CIC CNN), 30,283 (N-BaIoT MLP), and 8,907 (N-BaIoT CNN).

Block	Mean hours	Max hours	Sum hours
T1	1.774	1.980	49.686
T2	2.363	2.512	66.165
T5	0.542	0.585	15.182
T6	1.461	1.473	40.921

VI. DISCUSSION AND LIMITATIONS

The five-part evaluation is a way of reading the completed work. It was not one protocol written before every part. Parts 1, 2, and 4 were specified before their own results were used. Part 3, the sign-flip check, and Part 5, the data audits, were added later. Treating all five as if they had been fixed in advance would overstate the plan.

Part 1 separates trimming from client-size weighting. The three-seed CIC grid meets the descriptive trim-help and weight-hurt rules in all four model-by-α cells. The three-seed N-BaIoT device partition meets neither rule for either model. Those directional patterns can be placed side by side. They do not show that client construction caused the difference, because the dataset changes at the same time. Part 2 then supports only one component, T2 FA $-$ UW, and supports neither complete hypothesis. Part 4 supports none of its four tests. An apparent benefit of trimming or of size weighting in one processed corpus and one client construction did not become a general ranking.

Part 5 limits what those rankings can mean. Table 11 shows that the CIC $\alpha = 0.1$ partitions are far more unbalanced, in size and in label divergence, than the device partition. About 31.64% of CIC test rows and 59.72% of N-BaIoT test rows are exact copies of a stored training row, and every N-BaIoT test row comes from a device that also appears in training.

The official decision is the paired-t Holm family in Table 8. The exact sign-flip family in Table 9 is the sensitivity a reader should see at $n = 7$: after Holm correction it supports no comparison, and the family size makes a 0.05 exact decision unreachable. That sensitivity does not erase the preregistered paired-t result. It limits how strongly a seven-seed sign pattern can be read.

The three-seed tables estimate limited run-to-run variation. The seven-seed family includes no CIC $\alpha = 0.5$ cell. The nine device identities are the observed N-BaIoT device set in this corpus. The $20{,}000$-row cap changes natural client-size balance. The pooled-train scaler is not private per-device preprocessing. The per-class figure is a three-seed CIC display, not a seven-seed test and not a rare-class result. FedNova was not brought to a decision and remains deferred, so its accuracy on these blocks is unmeasured. SCAFFOLD was

not an arm. No malicious-client attack, secure aggregation, differential-privacy guarantee, wireless impairment, or physical deployment is evaluated. Full-scaler scores, including IID anchors for their original arms, stay in a separate lineage and supply no IID observation for unweighted FedAvg or for size-weighted trimmed mean. Study identifiers, approval dates, and checksums are in the supplementary note.

VII. CONCLUSION

The five parts have direct answers, and those answers keep the distinction between planned comparisons and later checks. In Part 1, the three-seed CIC directional pattern is not repeated on the N-BaIoT device partition, and that comparison does not isolate client construction because the dataset also changes. In Part 2, the preregistered seven-seed test supports one contrast: on the CIC convolutional network at $\alpha = 0.1$, size-weighted FedAvg finishes 10.7252 percentage points below unweighted FedAvg in Final macro-F1. The other 15 contrasts are inconclusive, and neither complete hypothesis is supported. In Part 3, the post-hoc sign-flip check supports none of those 16 contrasts after Holm correction. In Part 4, none of four Holm tests supports a difference between the trimmed mean and the coordinate-wise median. In Part 5, both pooled test sets contain exact copies of stored training rows, and the N-BaIoT test uses the same devices as training. Clients are simulated partitions. The apparent benefits of trimming and of client-size weighting depend on the dataset and on the client construction.

VIII. DATA AND REPRODUCIBILITY

The archive below stores the materials for the five-part evaluation: the two-by-two cells, the seven-seed cells used in Parts 2 and 3, the median cells used in Part 4, and the partition and matrix records used in Part 5. Training used Python 3.12.10, NumPy 2.3.5, pandas 3.0.1, scikit-learn 1.8.0, SciPy 1.17.1, and PyTorch 2.5.1 with CUDA 12.1 on a Quadro P3200. The new figures were drawn with Matplotlib 3.10.8. The primary local sources are the CIC confirmatory packet, the N-BaIoT device packet, the seven-seed packet, the 28 median cells, and the descriptive per-class table. Their manifests cover the written results and the processed caches. They do not independently hash every original source CSV. The public archive for this manuscript is https://github.com/sheryar827/fl-iot-trimming-weighting-archive. The supplement gives that address, the checksums of the two processed feature matrices, and the acquisition instructions for the public corpora. The result tables, cell records, manifests, and figure code are published in that repository. Those two matrices exceed the ordinary git file-size limit and are identified by checksum. The repository at https://github.com/sheryar827/fl-iot-botnet-nbaiot-cic-iot2023 is the earlier three-arm lineage, and its README labels it as that lineage. Headline comparisons there are not the results of this article. The record at https://doi.org/10.5281/zenodo.20456827 resolves to the immutable v1.0.0 snapshot https://doi.org/10.5281/zenodo.20456828, published on 30 May 2026. That snapshot is the three-arm study. It was not given a new version for this article, and it does not contain these packets.

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